14 **D**

$P = 1/3\ \rho \langle c^2 \rangle$

$1.00 \times 10^5 = 1/3\ (1.60) \langle c^2 \rangle$

$\langle c^2 \rangle = 1.86 \times 10^5\ m^2\ s^{-2}$

$C_{rms} = 433\ m\ s^{-1}$

15 **A**

At displacement = $\dfrac{A}{2}$, $u^2 = \omega^2 \left[A^2 - \left(\dfrac{A^2}{4} \right) \right]$

At displacement = 0, $v^2 = \omega^2 A^2$

Therefore $v = \dfrac{2}{\sqrt{3}}\ u$

16 **B** The oscillation experiences greater damping in oil than in water. Maximum amplitude